

**Section I****8 marks**

Use the multiple-choice answer sheet for Questions 1 – 8

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**QUESTION 1**

The number 0.003489 written in scientific notation to 2 significant figures is:

- A.  $3.489 \times 10^{-3}$       B.  $34.89 \times 10^{-4}$   
C.  $35 \times 10^{-4}$       D.  $3.5 \times 10^{-3}$

**QUESTION 2**

The expanded version of  $(2x - 1)^2$  is:

- A.  $4x^2 - 4x + 1$       B.  $4x^2 - 1$   
C.  $2x^2 + 1$       D.  $4x^2 - 4x - 1$

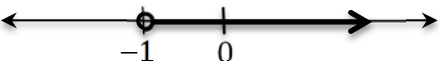
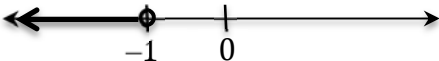
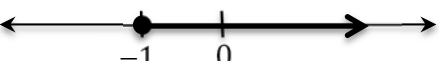
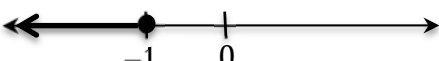
**QUESTION 3**

Expressed with a rational denominator,  $\frac{2}{3\sqrt{3}}$  would be written:

- A.  $\frac{\sqrt{6}}{3}$       B.  $\frac{2\sqrt{3}}{27}$   
C.  $\frac{2\sqrt{3}}{9}$       D.  $\frac{2\sqrt{3}}{3}$

**QUESTION 4**

The solution to the inequation  $1 - 2x < 3$  is represented on the number line as:

- A.       B.   
C.       D. 

**QUESTION 5**

$\left(-2\frac{2}{5}\right)^{-1}$  is the same as:

A.  $2\frac{2}{5}$

B.  $\frac{5}{12}$

C.  $-\frac{5}{4}$

D.  $-\frac{5}{12}$

**QUESTION 6**

$3 - |3 - 7|$  is equal to:

A. 7

B. -1

C. -7

D. 1

**QUESTION 7**

To complete the square on the expression  $x^2 - 16x$  you would need to add:

A. 4

B. 16

C. 64

D. 256

**QUESTION 8**

If  $T = \frac{m_1 - m_2}{1 - m_1 m_2}$  find the value of  $m_2$  when  $T = -4\frac{1}{2}$  and  $m_1 = -\frac{1}{2}$ :

A.  $1\frac{3}{13}$

B.  $-3\frac{1}{5}$

C.  $1\frac{3}{5}$

D.  $-1\frac{1}{15}$

**Section II****72 marks**

Start each question in a separate booklet for Questions 1 – 8

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**QUESTION 9** (10 Marks) **(Start a new page)** **Marks**

- (a) Convert  $0.0\dot{4}\dot{5}$  to a fraction in its simplest terms **2**
- (b) Write  $y^{\frac{3}{5}}$  without fractional indices **1**
- (c) Simplify
- (i)  $\sqrt{18}$  **1**
- (ii)  $2\sqrt{5} + 3 - 3\sqrt{5} + 1$  **1**
- (iii)  $3\sqrt{2} \times 5\sqrt{14}$  **2**
- (iv)  $\frac{5\sqrt{6}}{\sqrt{3}}$  **1**
- (d) Express  $\frac{\sqrt{2}}{\sqrt{5} - \sqrt{2}}$  with a rational denominator **2**

**QUESTION 10** (13 Marks) **(Start a new page)**

- (a) Simplify
- (i)  $14ab^2 \div 28ab$  **2**
- (ii)  $-2st^3 \times -5st^6$  **2**
- (b) Factorise fully
- (i)  $b^2 - 25$  **1**
- (ii)  $ab + 7a - 4b - 28$  **1**
- (iii)  $w^2 - 7w + 10$  **1**
- (iv)  $8x^2 + 31x + 21$  **2**
- (v)  $27t^3 + 125$  **1**
- (vi)  $2x^3 - 2x^2 - 2x + 2$  **3**

**QUESTION 11** (12 Marks) **(Start a new page)** **Marks**

(a) Simplify

(i)  $\frac{4a-2b}{4}$  **1**

(ii)  $\frac{(a+b)(a-b)}{3ab} \times \frac{9a}{(a-b)^2}$  **2**

(b) Express as a single fraction in simplest terms

(i)  $\frac{n}{3} - \frac{n}{4}$  **2**

(ii)  $\frac{2}{3ab} + \frac{1}{9a}$  **2**

(iii)  $\frac{y}{4-y^2} + \frac{y-1}{6+5y+y^2}$  **3**

(iv)  $\frac{a^3-b^3}{a-b} \div \frac{a^2+ab+b^2}{a+b} \div \frac{1}{a^2+2ab+b^2}$  **2**

**QUESTION 12** (11 Marks) **(Start a new page)**

Solve for x:

(a)  $\frac{2x-1}{3} - 5 = \frac{x}{6}$  **2**

(b)  $|x+2| \leq 3$  **3**

(c)  $|2x-1| = x+3$  **3**

(d)  $x^{\frac{2}{5}} = \frac{9}{4}$  **1**

(e)  $64^{x+3} = 4^x$  **2**

**QUESTION 13** (13 Marks) **(Start a new page)** **Marks**(a) Solve for  $x$  giving exact solutions:

(i)  $(x-4)^2 = 5$  **2**

(ii)  $3x(2x-1) > 0$  **2**

(iii)  $2x^2 + x = 3$  **2**

(iv)  $\frac{5x+7}{x-1} = 3x+2$  **2**

(b) Solve simultaneously:

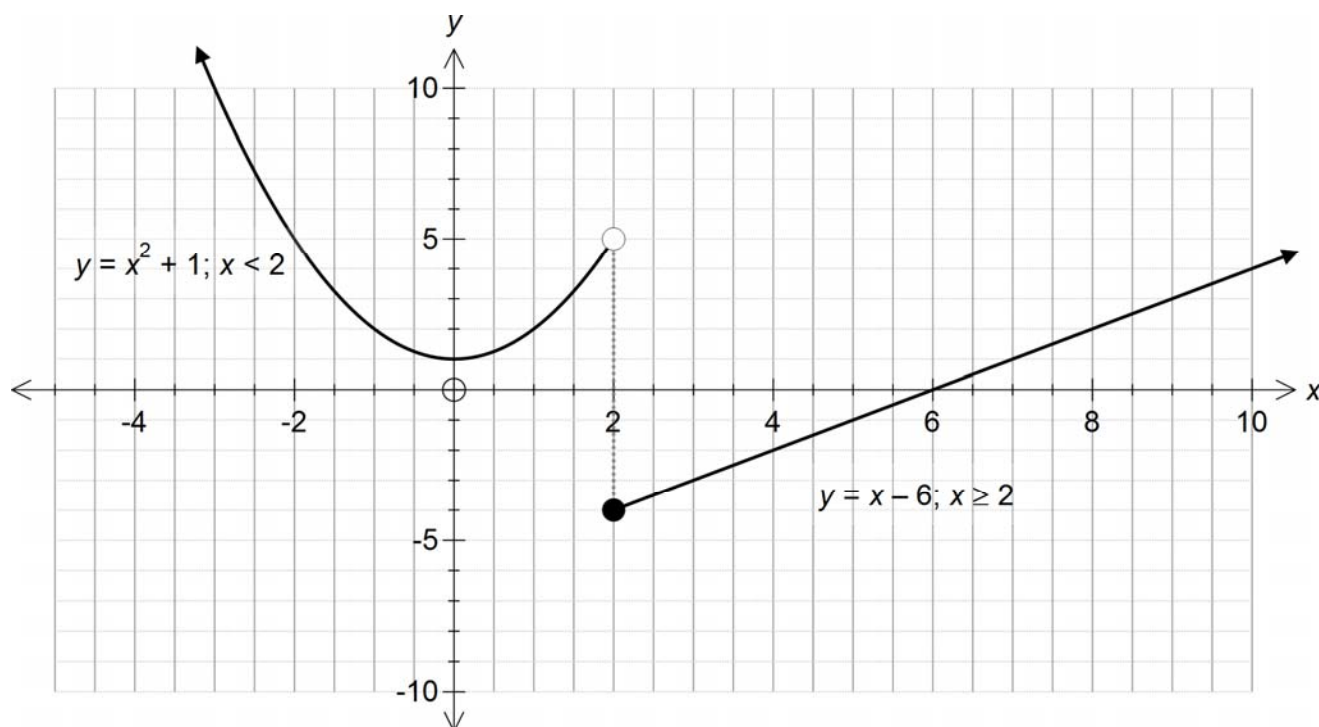
(i) 
$$\begin{aligned} 3x - y &= 5 \\ 5x + 3y &= -8 \end{aligned}$$
 **2**

(ii) 
$$\begin{aligned} x - y &= 1 \\ xy &= 2 \end{aligned}$$
 **3**

**QUESTION 14** (13 Marks) **(Start a new page)**(a) Is the following set of points a function?  $\{(1,1), (2,2), (3,5), (3,9)\}$  **1**(b) If  $f(x) = \begin{cases} x+1 & x \leq 3 \\ x^2 & x > 3 \end{cases}$ , find  $f(-3)$  **1**(c) Consider the function  $f(x) = (x-3)^2$ (i) Find the  $x$  and  $y$  intercepts. **2**(ii) Write down the domain and range clearly indicating which is which. **2**(d) **Show** whether the function given by  $f(x) = \frac{x}{x^2-1}$  is odd, even or neither. **2***Question 14 continues page 6.*

**QUESTION 14 continued.****Marks**

- (e) (i) Give the domain and range for the graph below. **2**  
(ii) Is the graph a function? **1**



- (f) Draw a neat graph of  $3x - 2y - 12 = 0$  clearly marking on any intercepts with the axes. **2**

**END OF EXAM**

2013 Solutions Unit 11.

- 1) D
- 2) A
- 3) C
- 4) A
- 5) D
- 6) B
- 7) C
- 8) B

Section 2.

Question 9.

a)  $0.\overline{045}$

let  $x = 0.0454545\dots$

$$100x = 4.545455$$

$$99x = 4.5$$

$$x = \frac{4.5}{99}$$

$$= \frac{45}{990}$$

$$= \frac{1}{22}$$

b)  $y^{3/5} = \sqrt[5]{y^3}$

c) i)  $\sqrt{18} = 3\sqrt{2}$

ii)  $2\sqrt{5} + 3 - 3\sqrt{5} + 1$   
 $= 4 - \sqrt{5}$

iv)  $3\sqrt{2} \times 5\sqrt{4}$   
 $= 15\sqrt{8}$   
 $= 30\sqrt{2}$

$$\text{iv) } \frac{5\sqrt{6}}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{5\sqrt{18}}{3}$$

$$= \frac{15\sqrt{2}}{3}$$

$$= 5\sqrt{2}$$

$$\text{d) } \frac{\sqrt{2}}{\sqrt{5}-\sqrt{2}} \times \frac{\sqrt{5}+\sqrt{2}}{\sqrt{5}+\sqrt{2}}$$

$$= \frac{\sqrt{10} + 2}{5 - 2}$$

$$= \frac{\sqrt{10} + 2}{3}$$

Question 10

$$\text{a) i) } 14ab^2 \div 28ab$$

$$= \frac{b}{2}$$

$$\frac{14ab^2}{28ab} = \frac{b}{2}$$

$$\text{ii) } -2st^3 \times -5st^6$$

$$= 10s^2t^9$$

$$\text{b) i) } b^2 - 25$$

$$= (b-5)(b+5)$$

$$\text{ii) } ab + 7a - 4b - 28$$

$$= a(b+7) - 4(b+7)$$

$$= (a-4)(b+7)$$

$$\text{iii) } w^2 - 7w + 10 \\ = (w-5)(w-2)$$

$$\text{iv) } 8x^2 + 31x + 21 \\ x = \frac{-31 \pm \sqrt{961 - 4(21)(8)}}{16} \\ = \frac{-31 \pm \sqrt{289}}{16} \\ = \frac{-13 \pm 17}{6}$$

$$x = -30/6 \text{ or } 4/6 \\ = -5 \text{ or } 2/3$$

$$\therefore 8x^2 + 31x + 21 = (x+5)(3x-2)$$

$$\text{v) } 27t^3 + 125 \\ = (3t+5)(9t^2 - 15t + 25)$$

$$\text{vi) } 2x^3 - 2x^2 - 2x + 2 \\ = 2(x^3 - x^2 - x + 1) \\ = 2(x^2(x-1) - 1(x-1)) \\ = 2(x^2-1)(x-1) \\ = 2(x+1)(x-1)(x-1) \\ = 2(x+1)(x-1)^2$$

Qn 11

$$\text{a) } \frac{4a-2b}{4}$$

$$= \frac{2(2a-b)}{4} \\ = \frac{2a-b}{2}$$

$$\text{a) ii) } \frac{(a+b)(a-b)}{\cancel{3ab}} \times \frac{\cancel{9a}}{(a-b)^2}$$

$$= \frac{3(a+b)}{b(a-b)}$$

$$\text{b) i) } \frac{n}{3} - \frac{n}{4}$$

$$= \frac{4n - 3n}{12}$$

$$= \frac{n}{12}$$

$$\text{ii) } \frac{2}{3ab} + \frac{1}{9a}$$

$$= \frac{6+b}{9ab}$$

$$\text{iii) } \frac{y}{4-y^2} + \frac{y-1}{6+5y+y^2}$$

$$= \frac{y}{(2-y)(2+y)} + \frac{y-1}{(y+3)(y+2)}$$

$$= \frac{y(y+3) + (y-1)(2-y)}{(2-y)(y+3)(y+2)}$$

$$= \frac{y^2 + 3y + 2y - y^2 - 2 + y}{(2-y)(y+3)(y+2)}$$

$$= \frac{6y-2}{(2-y)(y+3)(y+2)}$$

-5-

$$\text{iv) } \frac{a^3 - b^3}{a - b} \div \frac{a^2 + ab + b^2}{a + b} \div \frac{1}{a^2 + 2ab + b^2}$$

$$= \left( \frac{(a-b)(a^2 + ab + b^2)}{a-b} \div \frac{a^2 + ab + b^2}{a+b} \right) \times \frac{(a+b)^2}{1}$$

$$= \frac{a-b(a^2 + ab + b^2)}{a-b} \times \frac{a+b}{a^2 + ab + b^2} \times (a+b)^2$$

$$= (a+b)^3$$

Question 12.

$$\text{a) } \frac{2x-1}{3} - 5 = \frac{x}{6}$$

$$2(2x-1) - 30 = x$$

$$4x - 2 - 30 = x$$

$$3x = 32$$

$$x = \frac{32}{3}$$

$$\text{b) } |x+2| \leq 3$$

$$x+2 \leq 3$$

$$x \leq 1$$

$$\text{OR } -(x+2) \leq 3$$

$$-x-2 \leq 3$$

$$-x \leq 5$$

$$x \geq -5$$

$$\text{try } x=0$$

$$|0+2| \leq 3 \checkmark$$

$$-5 \leq x \leq 1$$

c)  $|2x-1| = x+3$

$2x-1 = x+3$

$x = 4$

or  $-(2x-1) = x+3$

$-2x+1 = x+3$

$-3x = 2$

$x = -2/3$

d try  $|2(4)-1| = 4+3$   
✓

$|2(-2/3)-1| = -2/3+3$

$|-8/3-1| = 2\frac{1}{3}$

$-1\frac{1}{3} = 7\frac{1}{3} \times$

∴ only  $x=4$

d)  $x^{2/5} = \frac{9}{4}$

$x^{2/5} = \left(\frac{3}{2}\right)^2$

$x = \left(\left(\frac{3}{2}\right)^2\right)^{5/2}$

$x = \frac{3^5}{2^5}$

e)  $64^{x+3} = 4^x$

$4^{3(x+3)} = 4^x$

$3(x+3) = x$

$3x+9 = x$

$2x = -9$

$x = -\frac{9}{2}$

Question 13.

$$\begin{aligned} \text{ai)} \quad (x-4)^2 &= 5 \\ \sqrt{(x-4)^2} &= \pm\sqrt{5} \\ x-4 &= \pm\sqrt{5} \\ x &= 4 \pm \sqrt{5} \end{aligned}$$

$$\text{ii)} \quad 3x(2x-1) > 0$$

$$\therefore x > 0 \quad \vee \quad 2x-1 > 0$$

$$2x > 1$$

$$x > \frac{1}{2}$$

$$\text{try } x = \frac{1}{4} \rightarrow -3/8 < 0$$

$$\therefore x > \frac{1}{2}$$

$$\text{iii)} \quad 2x^2 + x = 3$$

$$2x^2 + x - 3 = 0$$

$$\frac{2x^2 + x - 6}{2} = 0$$

$$\frac{(2x+6)(2x-1)}{2} = 0$$

$$(x+3)(2x-1) = 0$$

$$\text{iv)} \quad \frac{5x+7}{x-1} = 3x+2$$

$$5x+7 = (3x+2)(x-1)$$

$$5x+7 = 3x^2 - 3x + 2x - 2$$

$$5x+7 = 3x^2 - 5x - 2$$

$$3x^2 - 10x - 9 = 0$$

$$x = \frac{10 \pm \sqrt{100 - 4(-9)(3)}}{6}$$

$$= \frac{10 \pm \sqrt{208}}{6}$$

$$\therefore x = \frac{10 \pm 4\sqrt{13}}{6} = \frac{5 \pm 2\sqrt{13}}{3}$$

13 cont

$$\begin{aligned} \text{b) i)} \quad 3x - y &= 5 & \textcircled{1} \\ 5x + 3y &= -8 & \textcircled{2} \end{aligned}$$

From  $\textcircled{1}$   $y = 3x - 5$

$$5x + 3(3x - 5) = -8$$

$$5x + 9x - 15 = -8$$

$$14x = 7$$

$$x = \frac{1}{2}$$

$$\therefore x = \frac{1}{2}$$

$$\begin{aligned} y &= 3\left(\frac{1}{2}\right) - 5 \\ &= -\frac{7}{2} \end{aligned}$$

$$y = -\frac{7}{2}$$

check  $5\left(\frac{1}{2}\right) + 3\left(-\frac{7}{2}\right) = -8 \checkmark$

$$\text{ii) } x - y = 1 \quad \textcircled{1}$$

$$xy = 2 \quad \textcircled{2}$$

From  $\textcircled{1}$   $x = 1 + y$

Subst into  $\textcircled{2}$

$$(1+y)(y) = 2$$

$$y^2 + y = 2$$

$$y^2 + y - 2 = 0$$

$$(y-1)(y+2) = 0$$

$$\therefore y = 1 \text{ or } -2$$

$$x = 1 + y$$

$$x = 2 \text{ or } -1$$

$$x = 2 \text{ or } x = -1$$

$$y = 1$$

$$y = -2$$

check  $xy = 2$ :

$$(2 \times 1) (-1 \times -2) \& 2 - 1 = 1 \text{ or } -2 - 1 - -2 = 1 \checkmark$$